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CIA - 1

MOBILE COMPUTING - ITA0329

1. Given,

Metropolitan region, = 1200 km²

cell radius = 1.8 km

available Spectrum = 48 MHz = 48000 kHz

One full duplex channel = 200 kHz

$N_1 = 7$

$N_2 = 4$.

$Q = 4.5$

Area of Hexagonal cell = $\frac{3\sqrt{3}R^2}{2} \Rightarrow \frac{3\sqrt{3}(1.8)^2}{2} = 8.418\ km^2$

Number of cells = $\frac{\text{Total region}}{\text{Single cell region}} = \frac{1200}{8.418} = 142.5 \Rightarrow 143$ cells

Total available channel = $\frac{\text{available Spectrum}}{\text{full duplex channel}}$

$= \frac{48000}{200} = 240$.

With 15% reduction,

$\Rightarrow 240 \times \frac{85}{100} \Rightarrow 36 \Rightarrow 240 - 36 \Rightarrow 204$.

204 cells are efficient cells.

For N = 7.

$C_a = \frac{204}{7} = 29.14 \Rightarrow C = \frac{240}{7} = 34.28$.

for $N = 4$.

$$C_{re} = \frac{204}{4} = 51. \quad ; \quad C = \frac{240}{4} = 60$$

$N = 4$ is high but fair interference

So, $N = 7$ is best choice.

2. Given,

BSS handsoff = 120 ms

NSS hands off = 80. ms

for Authentication = 350 ms

for roaming registration = 400 ms.

Data loss = 512 kbps

No. of cells = 15 cells ; Speed = 90 km/h.

Ans

Normal Hands off delay = BSS handsoff + NSS handsoff

= 180 + 80 = 200 ms.

The Normal Hands off delay is 200 ms.

Total = Count - per handsoff

Total round = 15/200 = 3.0 s

Recovery = 1.2 s.

Romaing = Authentication + Reg.

= 2(350 + 250) = 1.2 s

Grand total = 5.4 s

But 3 s is limit

So it fails by 2.4 Seconds

To improve: ⇒ prediction handsoff

⇒ Make before break.

⇒ Buffering.

3. (i) Given,

BSC Manages = 12 BTS

TDMA slots = 8.

Sectors = 3

Transceivers = 4.

Users / Subscribers = 1200

Each making 1.2 calls per hour with average of 90 seconds.

Abis interface = 16 Kbps.

E1 = 2.048 Mbps.

TCH = 30 / sector.

Abis = 1.44 Mbps.

New Abis = 2.208 Mbps.

E1 is 1.104 Mbps

3. (ii). Given.

One-way Spectrum = 10 MHz = 10000 kHz.

FDMA use = 30 kHz

N of FDMA = 7.

TDMA User = 200 kHz width

N of TDMA = 4 at 8 Users per carrier.

CDMA = 1.25 MHz = 1250 kHz

CDMA with chip rate of = 1.226 Mcps.

CDMA Voice data rate = 9.6 Kbps.

FDMA = 333 total

So, 47/ cells

TDM = 50 Calls

= 400

So, 100 / cells

CDMA = 128

So, 128 / cells.

So, Best is CDMA.

because of Soft handoff

4. Given,

Total files = 240.

each Size = 19

Total file size = 240 × 19 = 4320 MB.

Time slots = 4.

Theoretical Data rate = 20 kbps.

No. of interruption = 8 of each 2.5 s.

Data delivery deadline = 12 hrs = 720 min = 43200 seconds.

$$\Rightarrow 4320 / \left(\frac{100-18}{100} \times 4\right)$$

$$= \frac{4320}{65.6} \Rightarrow 65.853\ s$$

If Protocol overhead reduced Capacity by 18%.

it takes 65.853 s to transfer the 240 images is the effective throughput.

The Connection experiences 8 interruptions each of 2.5 s is 20 seconds.

UMTS = 28 hrs

Total reduction is 95.48%.

Both fail dead at 12 hrs.

5. Given,

File Size = 120 GB.

No. of devices = 2000

GSM = 9.6 Kbps

GPRS = 64 Kbps with 10% retransmission traffic.

5G = 25 Mbps with 4% Protocol overhead.

Devices own 5G = 40%

Data Volume per device = $\frac{120,000,000}{9.6 \times 64 \times 25000 \times 0.04 + 64}$

$= \frac{120000000}{9.6 \times 70.4 \times 1000}$

60 Mb.

For

GSM = 18.89 hrs

GPRS = 2.29 hrs

5G = 24 Mbps / 20 Seconds

At 40% 800 devices, 8 batches. 5G is Best.

So, the Best Throughput is 5G.